

1

2

3

4

A

A

B

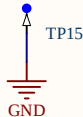
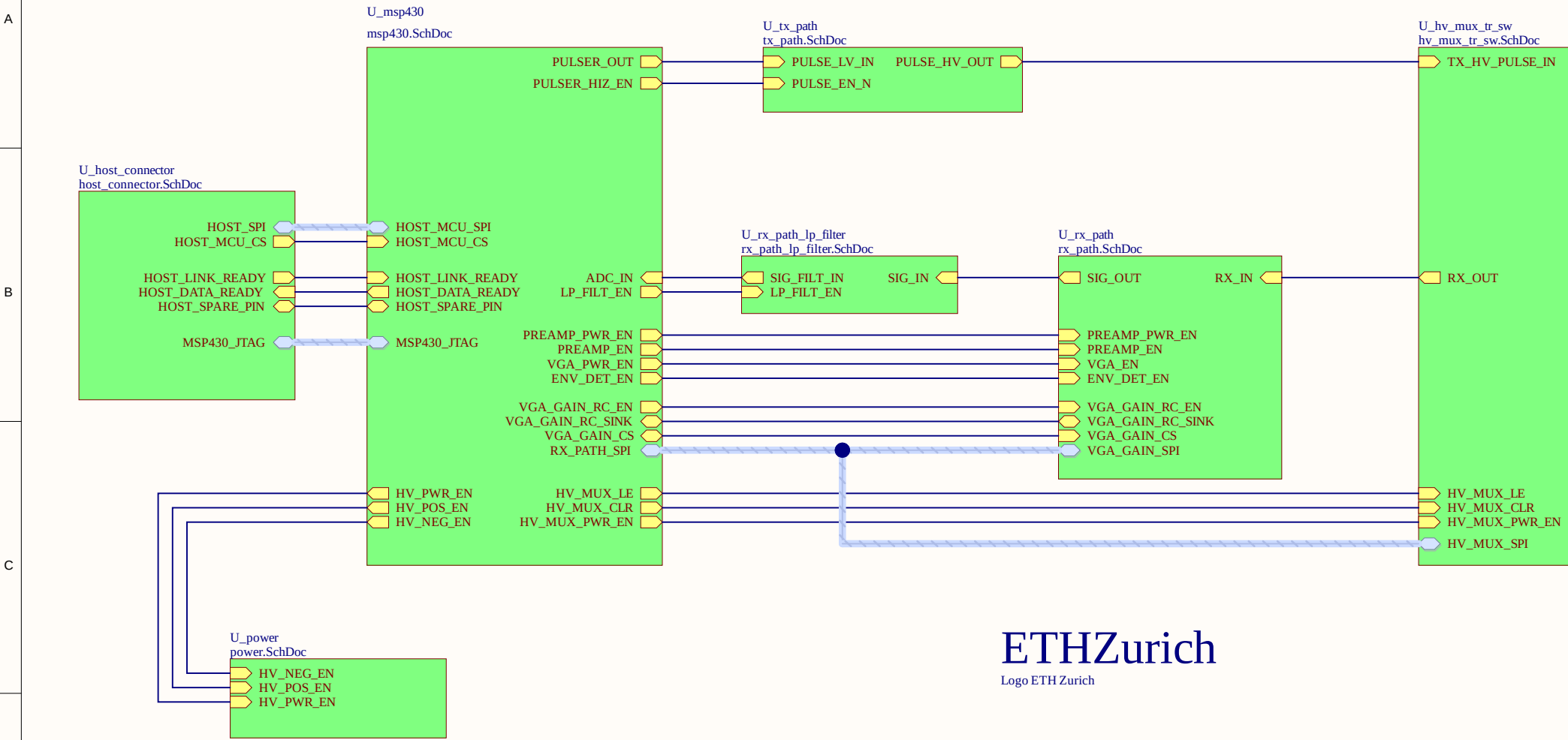
B

C

C

D

D



Mounting hole
BSL

ETHZurich
Logo ETH Zurich



Eidgenössische Technische Hochschule Zürich
wiss Federal Institute of Technology Zurich

Project:

BioGAP WULPUS-pro shield

rawing number: *

Rev: 1.0.0

Format:

Laboratory: *

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A4 Q

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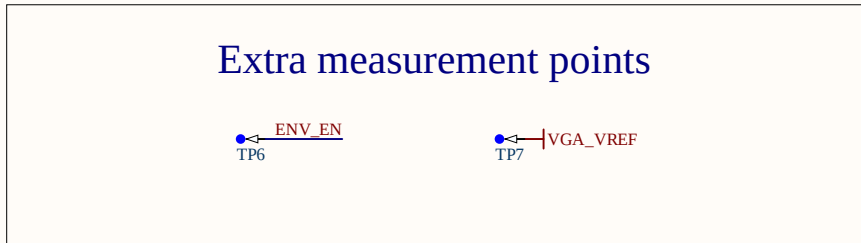
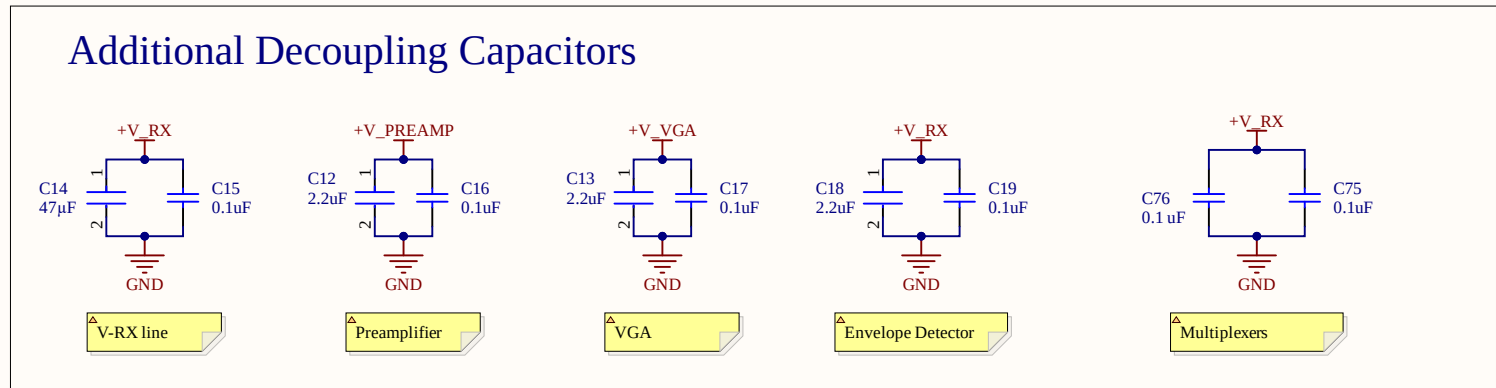
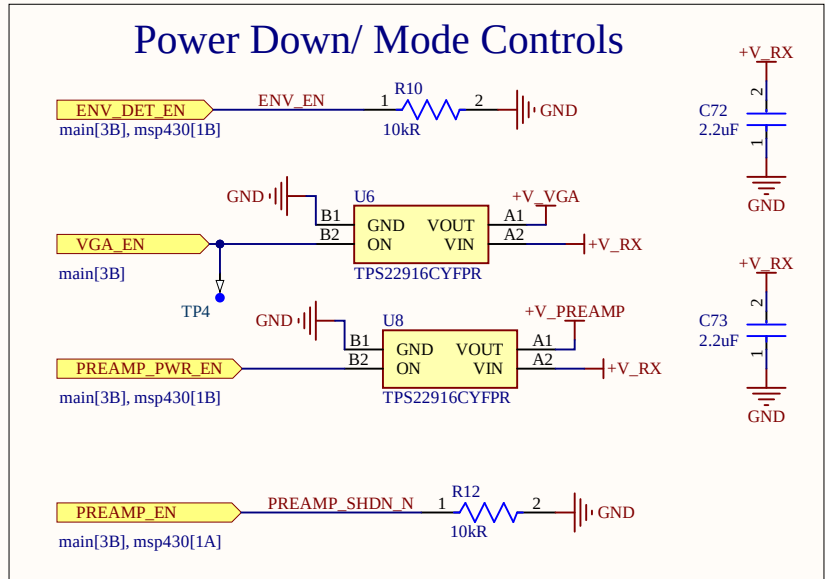
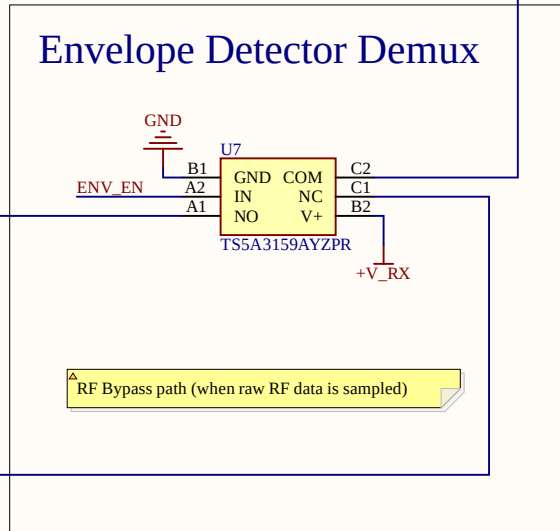
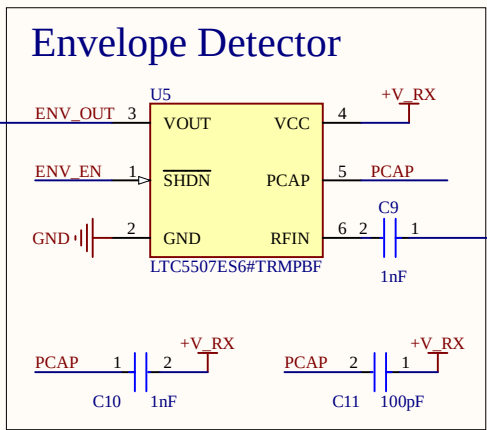
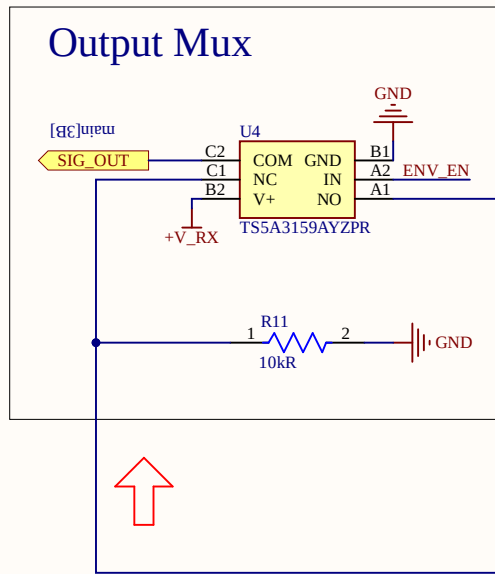
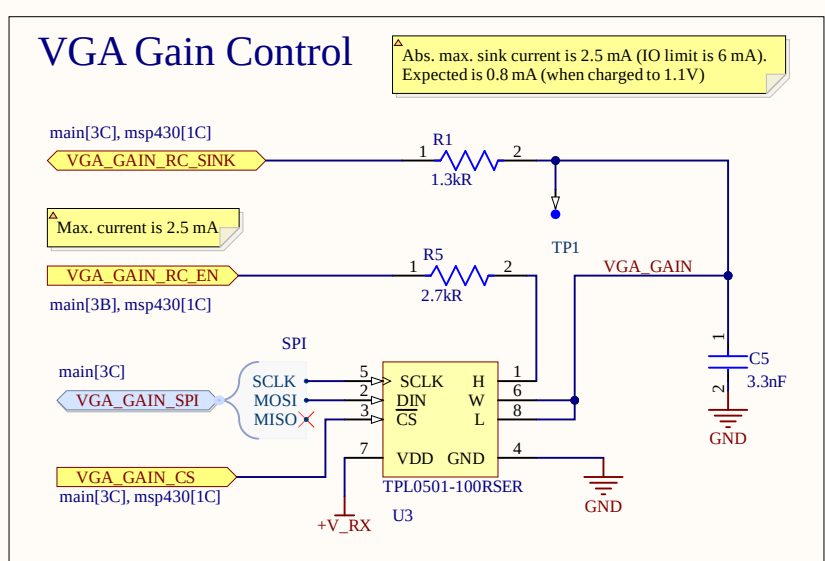
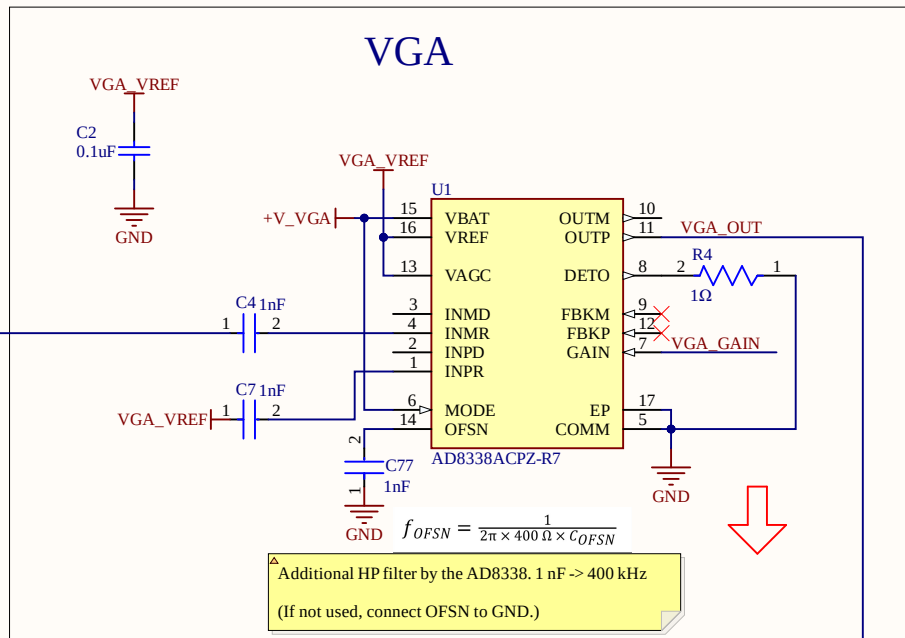
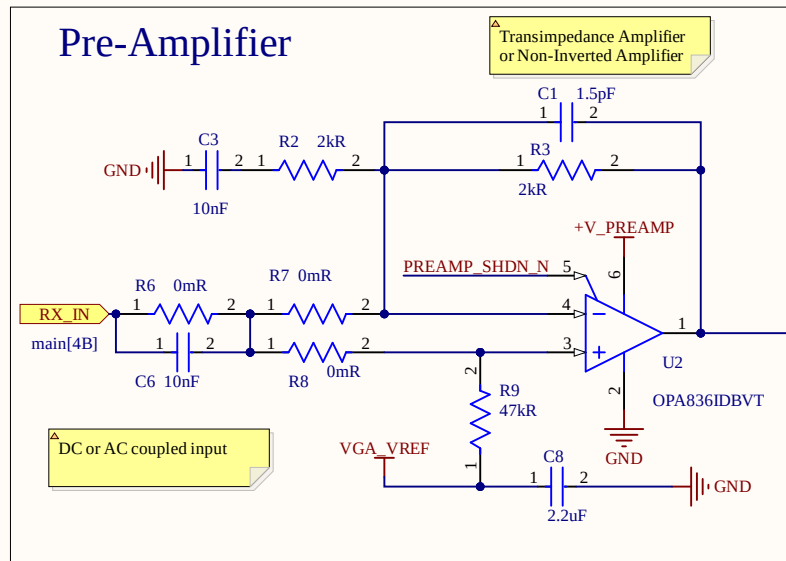
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1

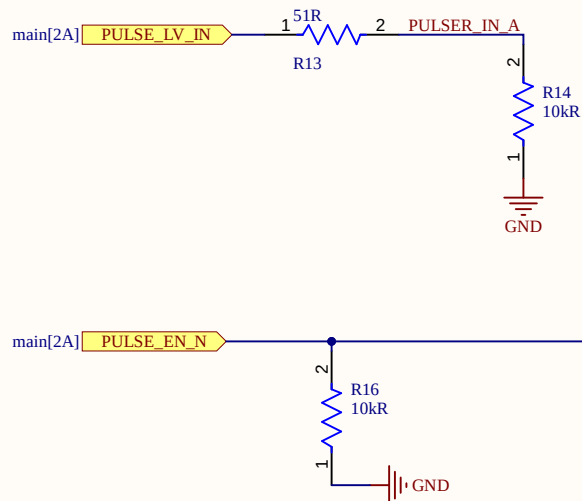
2

3

4



Input Matching



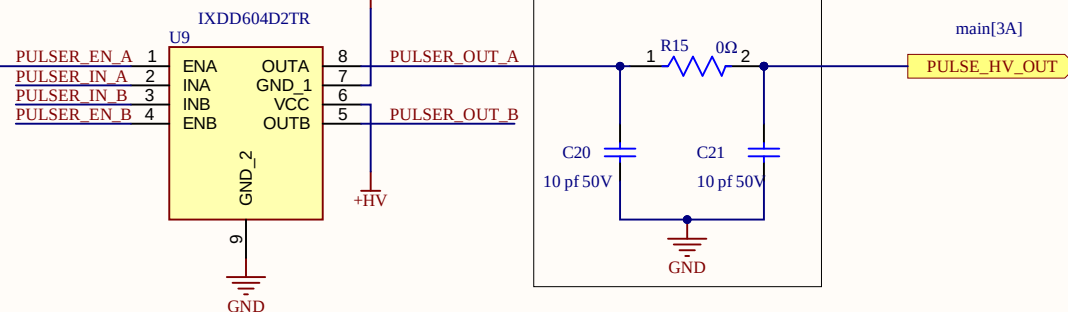
PULSE_EN_N drives EN pin through a transistor. It should be pulled to ground by default (enable state of the pulser)

PULSE_IN delivers square pulses 0-3.3V with the maximum frequency up to 10 MHz. It should be driven low if the PULSE_IN output is in high-Z state.

The transistor isolates MCU from the + HV supply, since EN_X pins are pulled up to VCC (HV) with 200k internally.

Pulser IXDD604D2TR has a max source resistance of 2.5 Ohm with typical rise time of 9 ns and fall time of 8 ns. Maximum source current is 9 A. Power supply current <1 uA from VCC.

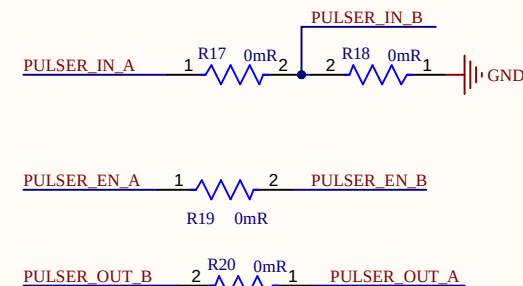
Optional Filter



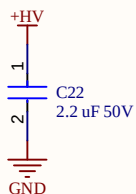
The second pulser channel (optional)

To connect the channel B, shorten pulser outputs PULSER_OUT_B and PULSER_OUT_A. Moreover, remove resistor that pulls PULSER_IN_B to ground and connect this line to PULSER_IN_A. Connect PULSER_EN_A to PULSER_EN_B

When using a single pulser channel A, disconnect all three signals (IN, EN and OUT) between channels A and B. Moreover, pull PULSER_IN_B to ground for the lowest power consumption.



Decoupling Capacitors



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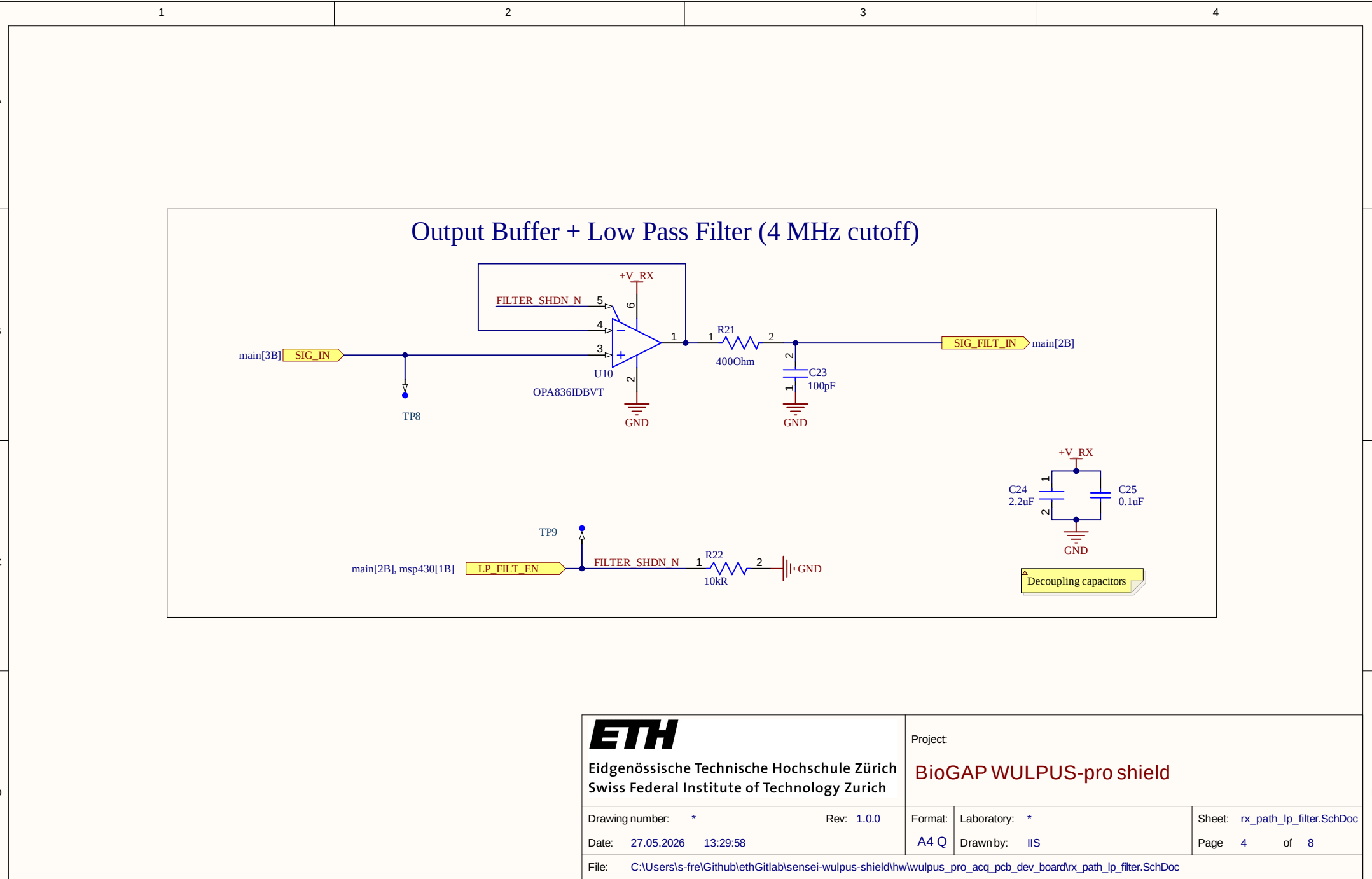
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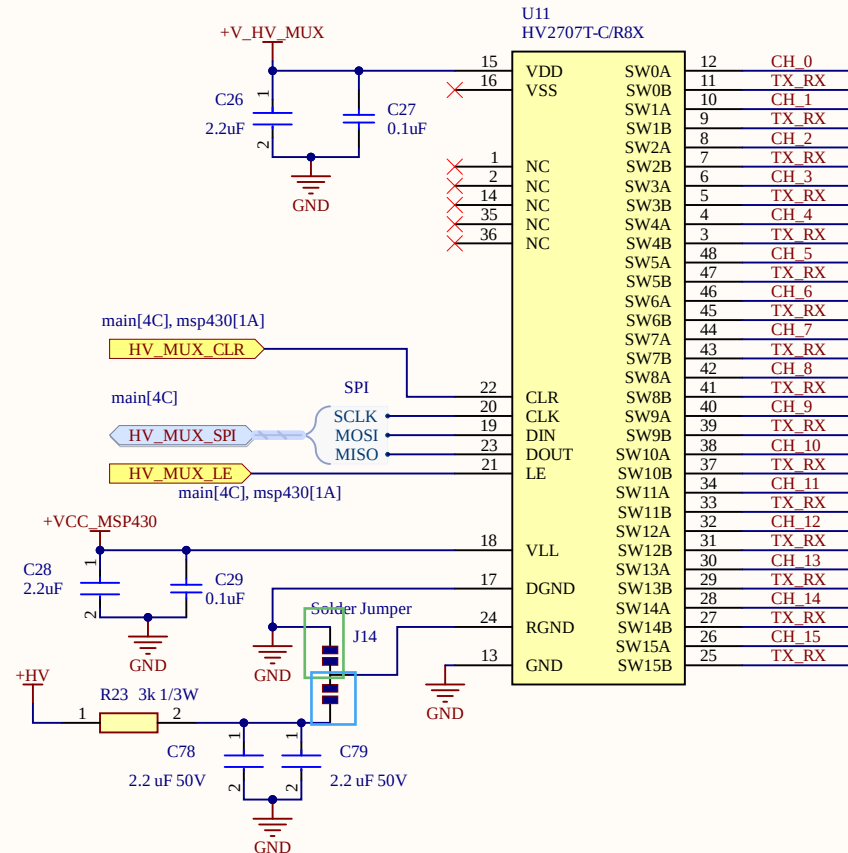
Drawn by: IIS

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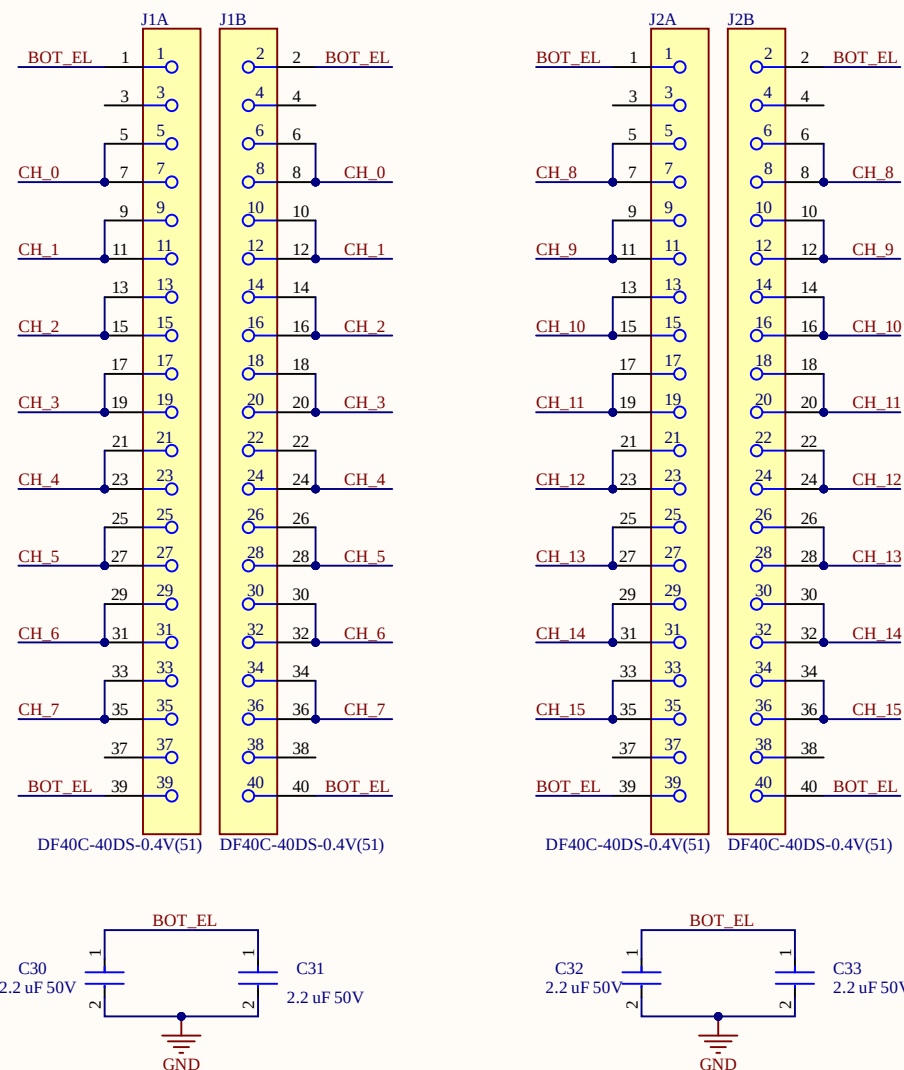
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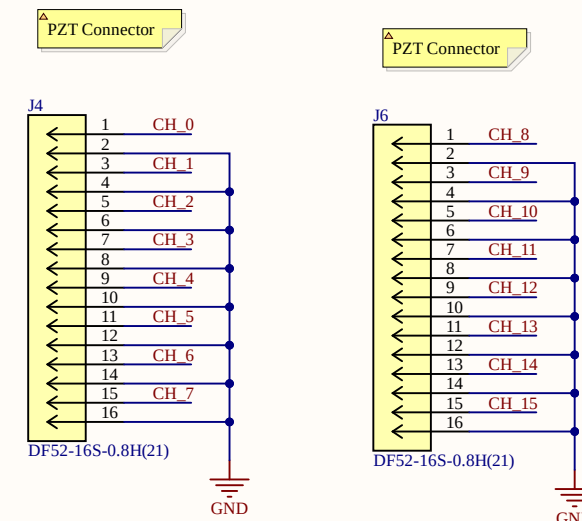
High-Voltage Multiplexer



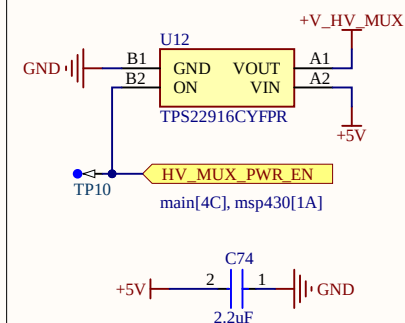
Transducer Connectors



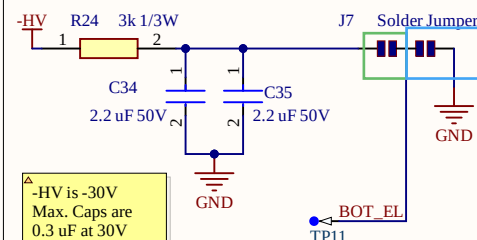
Board-to-wire Connectors



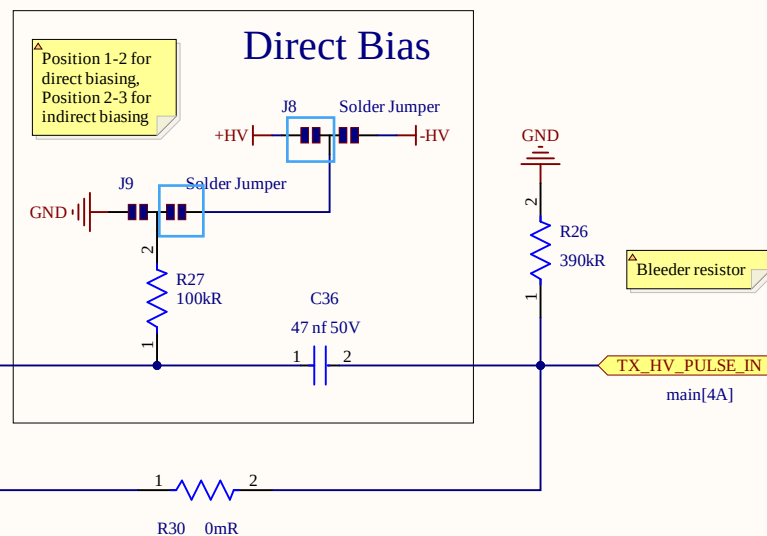
Power Gating+ 5V



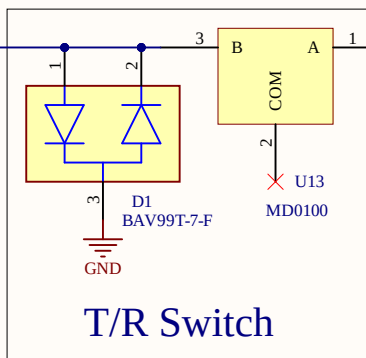
Indirect Bias



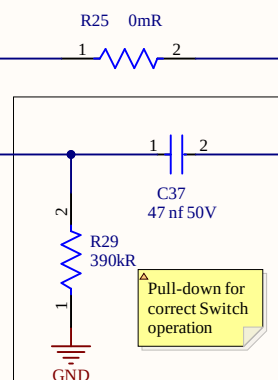
Direct Bias



T/R Switch



AC coupling



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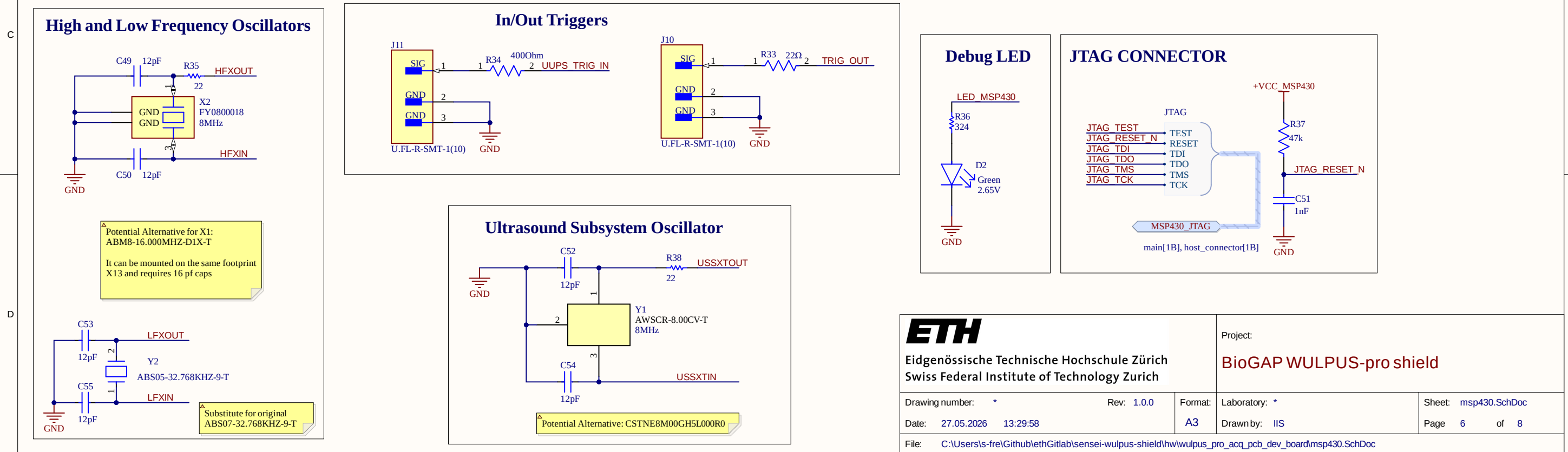
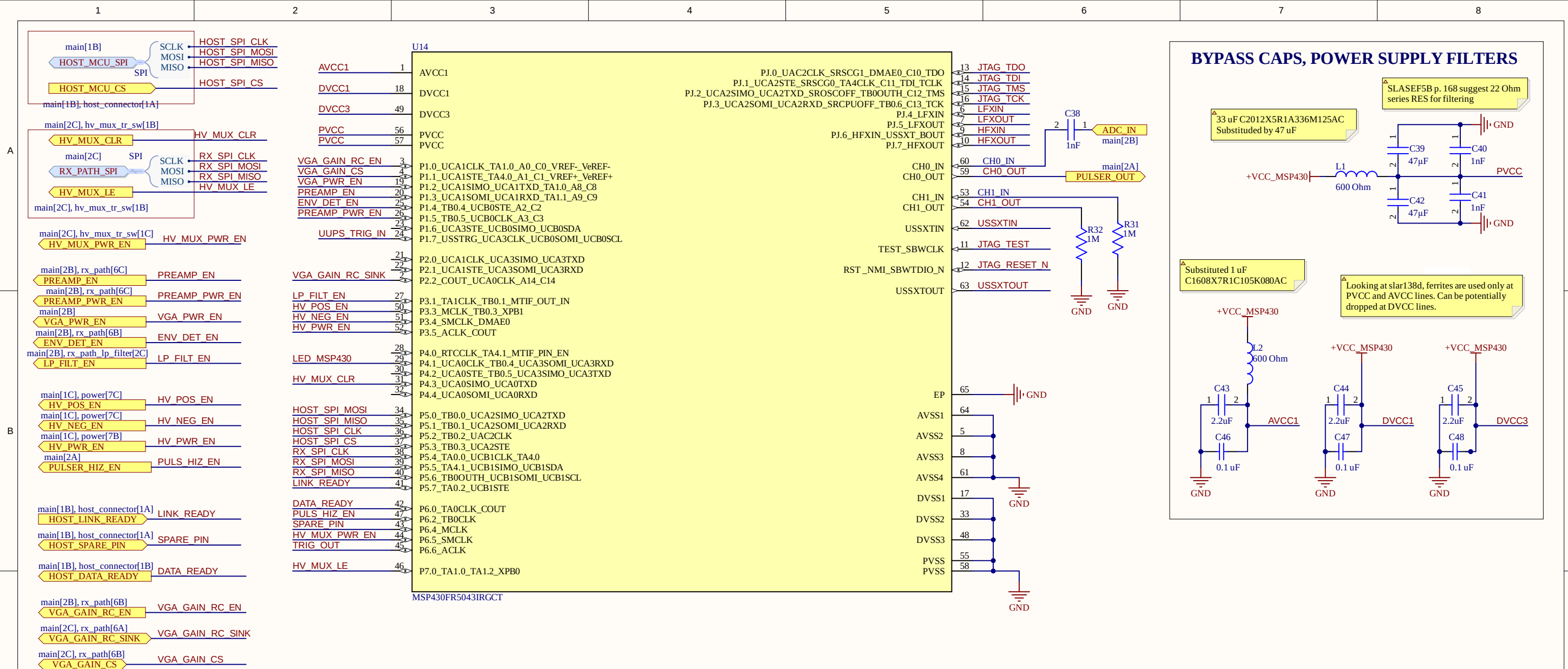
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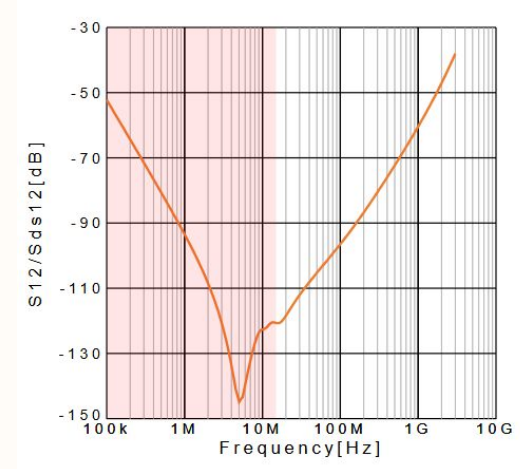
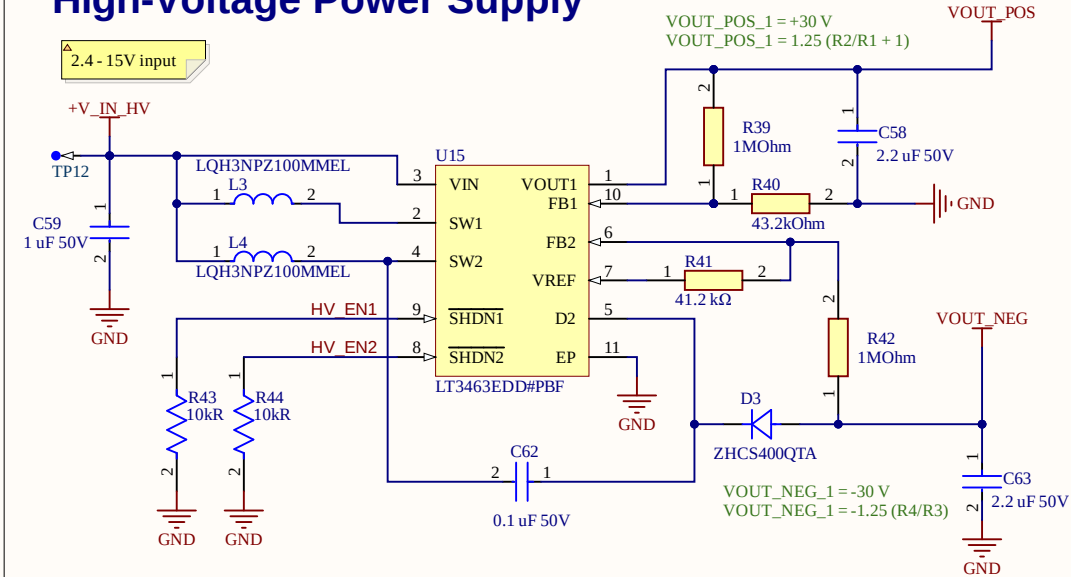
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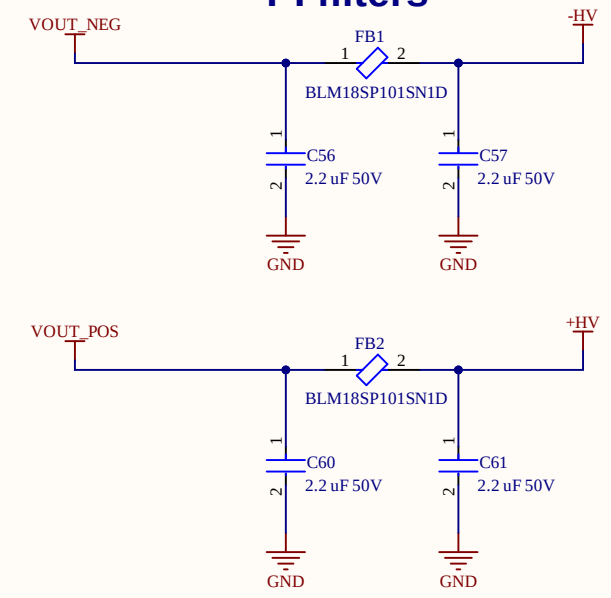


High-Voltage Power Supply

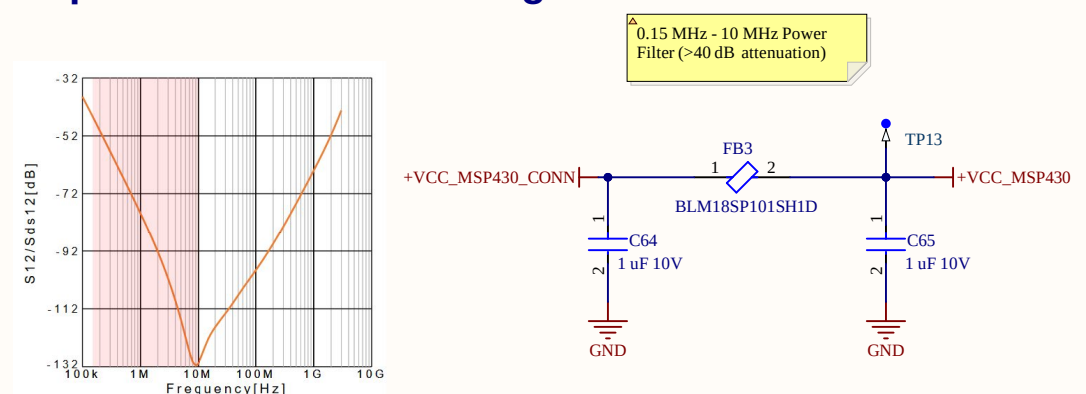


0.1 MHz - 10 MHz Power Filter (>50 dB attenuation). Simulation without effective capacitance.

PI filters



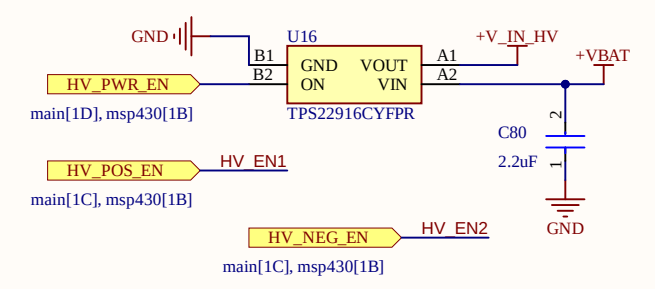
Input Filter for MCU voltage



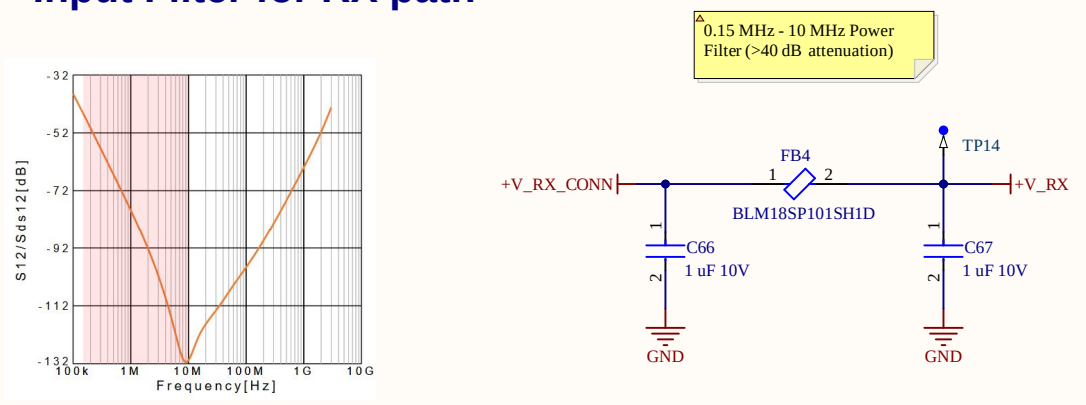
0.15 MHz - 10 MHz Power Filter (>40 dB attenuation)

Power Tree (No Load Switches)

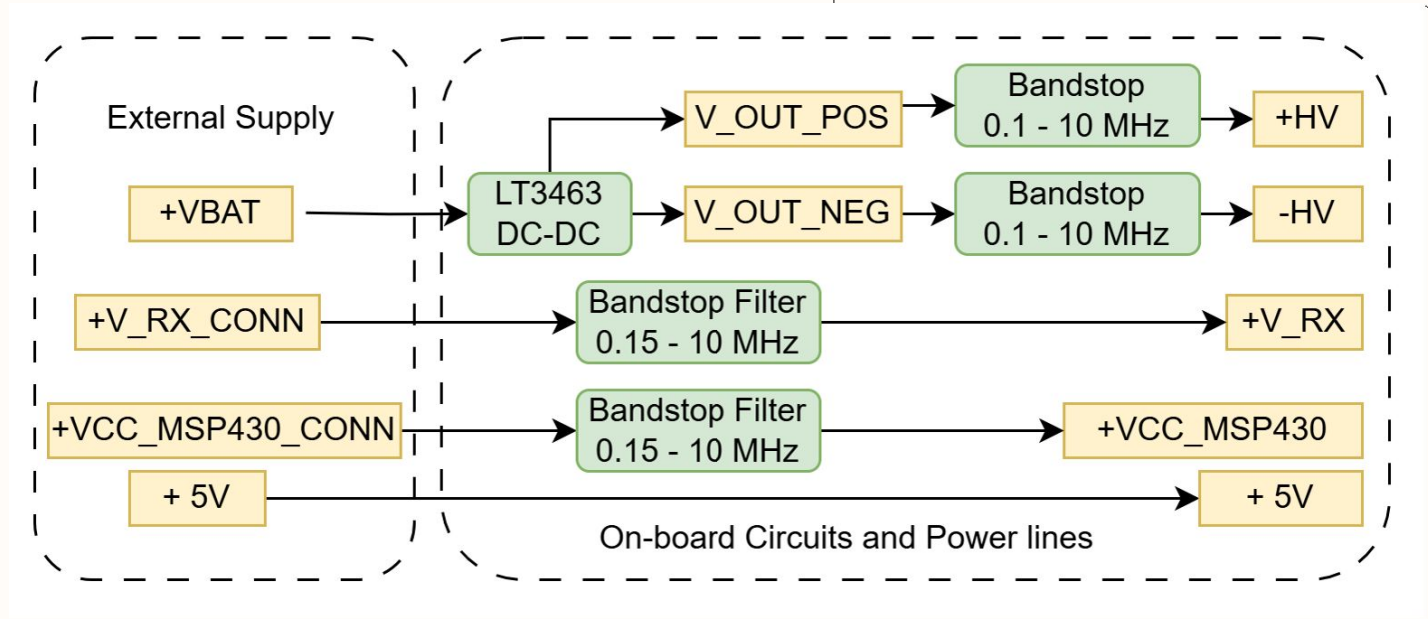
Power Switch

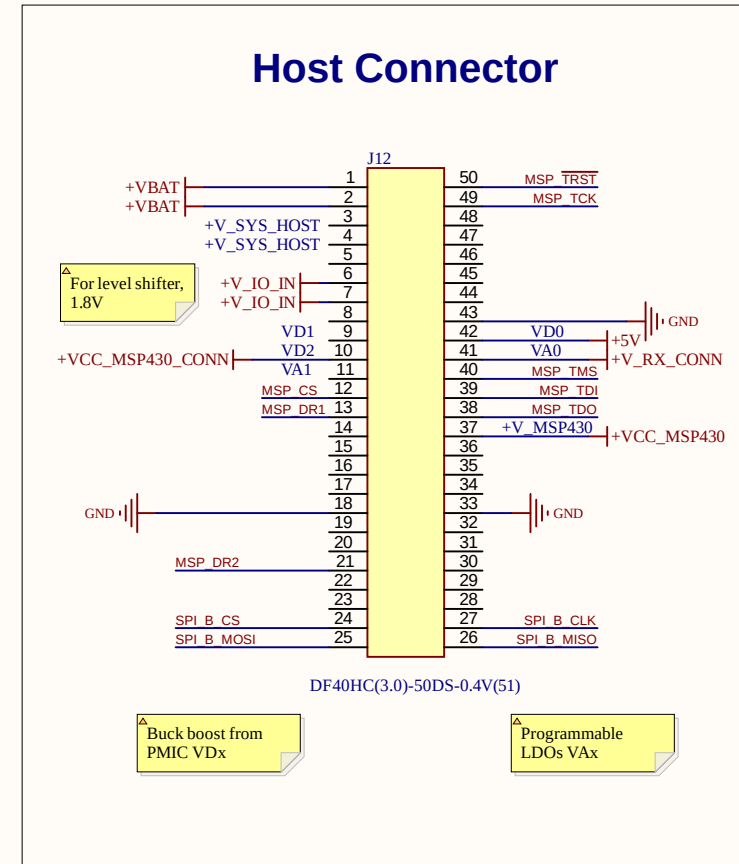
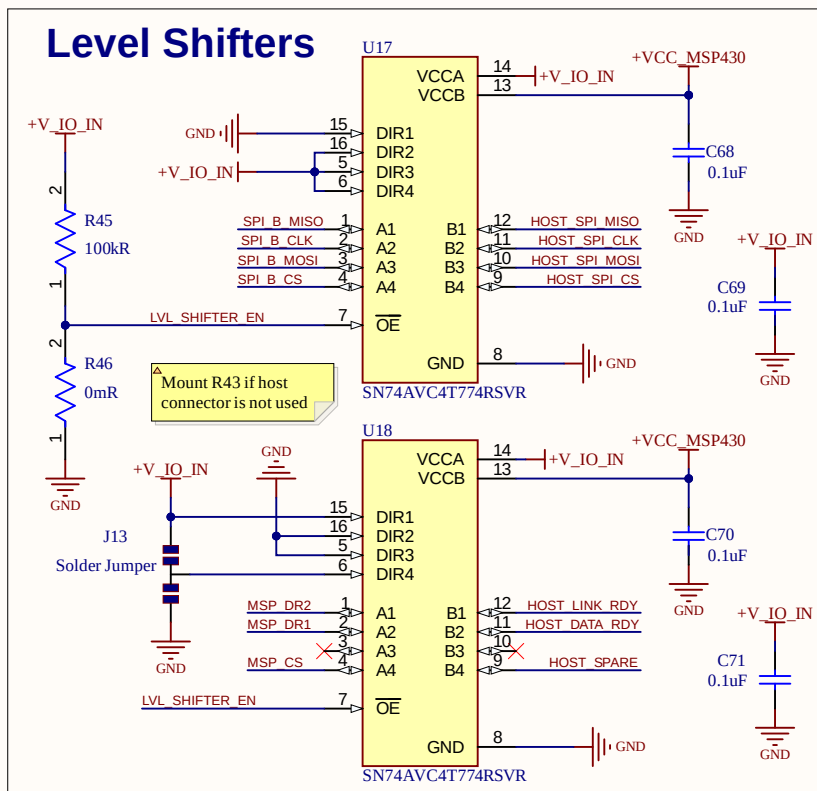
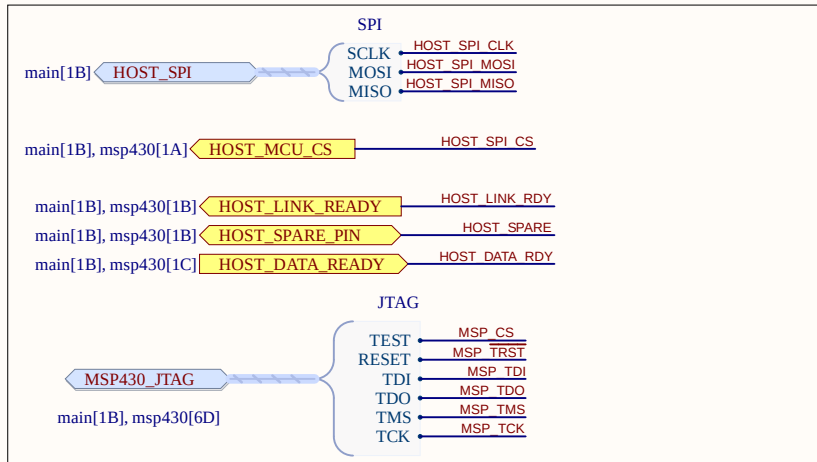


Input Filter for RX path



0.15 MHz - 10 MHz Power Filter (>40 dB attenuation)





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